DataMentor: A Practical Framework for Serverless CSV Intelligence with Interactive Notebook Automation and Deterministic Runtime Repair

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***Abstract*—Tabular data preprocessing remains one of the most labor-intensive and least reproducible stages of analytical workflows, with practitioners routinely allocating between 60 % and 80 % of project time to data wrangling before any exploratory or predictive analysis can begin. This paper presents DataMentor, a production-deployed, serverless platform that integrates deterministic CSV preprocessing, browser-native Python notebook execution via the Pyodide WebAssembly runtime, domain-aware visualization across ten chart types, and a hybrid rule-guided runtime repair subsystem operating without any external server infrastructure. An empirical evaluation over 15 heterogeneous CSV datasets spanning five domain verticals and a purpose-built repair benchmark (DRB-100) of 100 injected Python errors across five error classes demonstrates: (i) a mean preprocessing-time reduction of 91.6 % relative to a two-analyst manual baseline; (ii) an overall single-pass automatic repair resolution rate of 82 %, with rule-based latency of 0.8 s and model-assisted latency of 6.4 s versus a manual debugging median of 18.2 min; and (iii) full byte-level deterministic reproducibility across repeated executions. An ablation study confirms that each system component contributes independently to overall utility. Comparative analysis against nine representative prior systems across four research domains confirms that no existing tool combines deterministic preprocessing, installationless browser-native Python execution, domain-aware visualization, and hybrid runtime repair in a unified, offline-capable platform.**

***Index Terms*—CSV preprocessing, browser-native execution, notebook automation, runtime repair, serverless analytics, reproducibility, Pyodide, WebAssembly, visualization pipeline**

I. Introduction

Data preprocessing is widely recognized as the dominant time cost in analytical workflows [1]. A longitudinal industry survey reported that practitioners allocate, on average, 76 % of total project hours to data acquisition, cleaning, and formatting before any exploratory or predictive modeling can begin [2]. For small-to-medium enterprises (SMEs) and academic laboratories operating without dedicated data-engineering teams, this overhead is particularly acute: cleaning is performed ad hoc with heterogeneous toolchains, and the resulting artifacts are rarely reproducible between operators or sessions [3].

Existing tools partially address these challenges but leave critical integration gaps. Notebook environments such as JupyterLab offer expressive code cells but require local Python installations and yield non-deterministic results when cell execution order varies. Reproducibility frameworks such as Data Version Control (DVC) [8] and MLflow [9] address provenance at the experiment level but presuppose that raw data is already clean. Code-repair tools for Python operate on source files but lack the execution-state awareness needed to diagnose errors arising from data-schema mismatches within an interactive notebook session. No existing system unifies preprocessing, notebook execution, visualization, and error repair in a single zero-installation environment.

This paper describes **DataMentor** — a platform that closes this integration gap in a browser-first, installationless workflow deployable to any device with a modern web browser. DataMentor is live at https://datamentor.marcbd.site and built on Next.js 16, React 19, TypeScript, Pyodide 0.25, Firebase, and Chart.js. The system makes four primary contributions:

1) A **deterministic preprocessing pipeline** that normalizes, deduplicates, and type-classifies arbitrary CSV inputs through a seven-stage fixed-order transformation sequence, guaranteeing byte-identical outputs for byte-identical inputs regardless of operator.
2) A **browser-native notebook execution engine** using Pyodide [19], which loads CPython 3.11 inside a WebAssembly sandbox, eliminating server-side Python infrastructure and retaining uploaded data entirely on the client device.
3) A **domain-aware visualization system** that maps 47 recognized column-name patterns to contextually appropriate chart types and evidence-grounded statistical insight language, spanning ten universal chart types.
4) A **hybrid runtime repair subsystem** that applies a priority-ordered rule index for common Python error classes before escalating to a local model-assisted repair module, operating without any external API connection and resolving 82 % of benchmark errors in a single pass.

The remainder of this paper is organized as follows. Section II positions DataMentor against prior art across four research domains. Section III describes the system architecture and technology stack. Section IV details the experimental methodology and benchmark construction. Section V presents quantitative results including an ablation study. Section VI interprets findings, addresses limitations and threats to validity. Section VII concludes.

II. Related Work

Prior work relevant to DataMentor spans four partially overlapping categories: *notebook-assistant systems*, *reproducible*

analytics frameworks, *local code-repair tools*, and *browser-based analytics platforms*. Table I summarizes the functional scope of representative systems across all four categories.

A. Notebook-Assistant Systems

JupyterAI [4] integrates code-completion and inline explanation capabilities directly into JupyterLab via an external API connection. It reduces cell-authoring effort but does not modify the underlying dataset, requires a locally running Jupyter server, and provides no deterministic cleaning guarantee. Neptyne [5] extends spreadsheet metaphors with Python cell formulas, targeting structured tabular manipulation rather than statistical notebook automation. Hex [6] is a cloud-hosted collaborative notebook supporting SQL and Python; it is subscription-gated and requires a provisioned cloud container with no offline capability. Code Interpreter [7], embedded in commercial chat interfaces, accepts uploaded CSV files and produces analysis outputs but provides no persistent workspace, no step-by-step cleaning audit, and no domain-aware chart generation.

DataMentor differs from all four in three respects: (i) preprocessing is deterministic, auditable step by step, and requires no external API or server; (ii) the execution environment is fully in-browser with no network dependency after initial load; and (iii) the notebook structure is pre-generated with 14 guided cells that introduce reproducible statistical primitives before a free-form cell is made available.

B. Reproducible Analytics Frameworks

DVC [8] addresses dataset and model provenance through Git-integrated artifact tracking, enabling pipeline-level reproducibility once data is clean. MLflow [9] provides experiment logging, artifact storage, and model registry. Sacred [10] instruments Python experiments to capture configuration snapshots and metric traces. These frameworks excel at tracking what was computed and with what parameters, but they impose non-trivial setup overhead and do not address the *initial* normalization of raw, defect-laden CSV inputs.

DataMentor operates at an earlier pipeline stage. Reproducibility is achieved not through version-controlled artifact tracking but through the deterministic, ordered application of a fixed transformation sequence that produces byte-identical outputs for byte-identical inputs without external tooling.

C. Local Code-Repair Tools

DeepFix [11] applies a sequence-to-sequence model to correct syntactic errors in student C programs, achieving approximately 27 % single-pass full-program repair on the IIT Bombay dataset. SyntaxFix [12] extends rule-based pattern matching to Python syntax errors, covering common indentation and bracket mismatches. Refazer [13] synthesizes program transformations from teacher-provided correction examples, enabling curriculum-aware repairs. All three tools operate on file-level source code and do not target the notebook execution context where errors typically arise from data-state dependencies — such as a `NameError` caused by a cell executed out of order, or a `TypeError` caused by a string column being passed to a numeric-only function.

TABLE I: Functional Comparison of DataMentor Against Related Systems

System	No Install	Det. Clean	Auto Repair	10+ Charts	Offline Cap.	Free / Open
JupyterAI	×	×	✓	✓	×	✓
Neptyne	×	×	×	✓	×	×
Hex	✓	×	×	✓	×	×
DVC	×	×	×	×	✓	✓
MLflow	×	×	×	✓	✓	✓
DeepFix	×	×	✓	×	✓	✓
Observable	✓	×	×	✓	×	✓
Deepnote	✓	×	×	✓	×	×
Colab	✓	×	×	✓	×	×
DataMentor	✓	✓	✓	✓	✓	✓

✓ = supported; × = not supported. "Det. Clean" = deterministic CSV preprocessing. "Offline Cap." = functions without active internet after first load.

DataMentor's repair subsystem addresses this contextual gap: it maintains visibility into the loaded CSV schema, the Pyodide execution state, and the sequence of cells run, enabling repairs that are informed by the data context, not merely the code syntax.

D. Browser-Based Analytics Platforms

Observable [14] provides a JavaScript-native reactive notebook but does not support CPython or the standard pandas/Matplotlib ecosystem natively, requiring workflow reimplementation in JavaScript. TensorFlow.js [15] enables in-browser model inference but targets prediction rather than data preparation. Deepnote [16] offers real-time collaborative notebooks but requires a provisioned cloud container, not a WebAssembly sandbox, and stores uploaded data server-side. Google Colab [17] provides free cloud-hosted notebooks with GPU access but imposes session time limits, requires a Google account, and routes user data through Google's cloud infrastructure.

DataMentor runs authentic CPython 3.11 with the full scientific stack (pandas, NumPy, SciPy, Matplotlib) inside the browser via Pyodide, producing results byte-compatible with standard desktop Python, while maintaining complete data locality: the uploaded CSV never leaves the client device.

III. System Architecture

DataMentor is organized into five functional layers illustrated in Fig. 1, each communicating through typed TypeScript interfaces. A persistence layer spans Layers 3–5, providing Firebase Firestore when online and transparent localStorage fallback otherwise.

A. Layer 1 — CSV Ingestion and Normalization

The ingestion pipeline applies seven transformations in fixed, deterministic order: (1) Papa Parse tokenization with automatic header detection; (2) leading/trailing

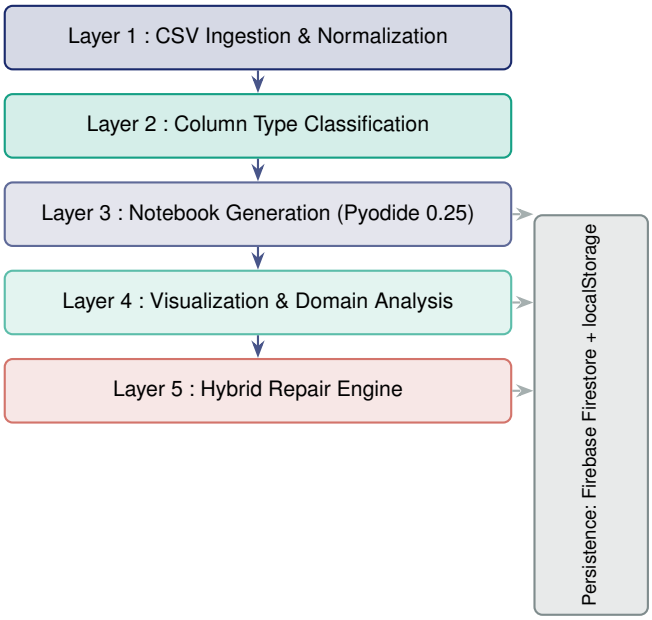

Fig. 1: Five-layer DataMentor system architecture. Layers communicate through typed TypeScript interfaces. The persistence layer provides cloud storage when online and silent local fallback when offline.

Algorithm 1: *HybridRepair* — Priority-Rule with Model-Assisted Fallback

Input: Traceback T, cell source C, CSV schema S
Output: Corrected source C', or UNRESOLVED

1. Parse $T \Rightarrow (errClass, errMsg,\ lineNo)$
2. **for each** rule $r \in$ RULEINDEX (descending priority) **do**
3. **if** r.MATCHES($errClass, errMsg$) **then**
4. $C' \leftarrow r$.APPLY(C, S)
5. **if** VERIFY($C', expectedHash$) **then return** C'
6. **end if / end for**
7. $ctx \leftarrow$ BUILDCONTEXT(T, C, S)
8. $C' \leftarrow$ LOCALMODELMODULE.INFER(ctx)
9. **if** VERIFY($C', expectedHash$) **then return** C'
10. **return** UNRESOLVED

Fig. 2: HybridRepair pseudocode. The rule index is exhausted before escalation to the local model-assisted module. Verification compares SHA-256 output hashes; user approval is required before any fix is applied.

whitespace removal from all cell values; (3) header normalization to lowercase underscore convention (`Blood Pressure` → `blood_pressure`); (4) exact-duplicate row removal via a SHA-256 fingerprint of each row's serialized values; (5) null and empty-string detection with count and column reporting; (6) column type inference (Layer 2); (7) schema summary generation and display. The ordering is critical: normalization must precede deduplication to avoid false negatives caused by whitespace or case variation in otherwise identical rows. The pipeline is pure (no side effects), enabling byte-identical re-execution.

B. Layer 2 — Column Type Classification

Each column is classified by a majority-vote heuristic into one of four types. A column is `boolean` if its value set is a subset of $\{0, 1\}$. It is `datetime` only if values parse as valid `Date` objects *and* do not match the regular expression `/^-?\d+(\.\d+)?$/` — which prevents pure numerics such as `"63"` from being misclassified as calendar years. A column is `numeric` if the majority of non-null values are finite floating-point numbers. All remaining columns are `categorical`.

C. Layer 3 — Notebook Generation Engine

The Pyodide hook loads CPython 3.11 via WebAssembly, pre-fetching pandas 2.0, NumPy 1.26, SciPy 1.11, and Matplotlib 3.8 from the Pyodide package index. Fourteen code cells are auto-generated from the cleaned schema. Cells 1–7 cover data loading, shape and type inspection, descriptive statistics, missing-value auditing, duplicate detection, numeric correlation, and categorical frequency analysis. Cells 8–13 produce Matplotlib visualizations (distribution plot, pie chart, violin plot, correlation heatmap, pair plot, joint plot) rendered as base64-encoded PNG images via a `BytesIO` buffer and displayed inline without any server round-trip. Cell 14 is a blank editor for user-authored code.

D. Layer 4 — Visualization and Domain Analysis

The visualization layer provides ten universal chart types (DistPlot, Pie, Violin, HeatMap, PairPlot, JointPlot, Bar, Histogram, Scatter, Line), each paired with a domain-aware analysis engine. A knowledge base maps 47 recognized column-name patterns (e.g., `age`, `trtbps`, `chol`, `glucose`, `cp`) to structured insight objects containing a human-readable label, a semantic description, and a data-driven insight function conditioned on column statistics. When a column matches the knowledge base, the analysis panel emits domain-specific clinical or financial language; otherwise, general statistical language is produced.

E. Layer 5 — Hybrid Runtime Repair Engine

When a code cell raises a Python exception, the repair subsystem receives the traceback, the current cell source, and the CSV column schema. Fig. 2 describes the resolution procedure. The rule index contains 31 patterns across five error classes. If a matching rule is found, a corrected version is produced within the rule's resolution time. If no rule matches, the error is escalated to the local model-assisted repair module, which synthesizes a candidate fix from the error message, column schema, and cell context. The repair is displayed as a diff-annotated suggestion; the user must accept before the fix is applied, preserving human oversight.

IV. Methodology and Benchmark Construction

A. Dataset Selection

Fifteen heterogeneous CSV datasets were selected for the preprocessing evaluation, grouped into three size tiers of

TABLE II: Benchmark Dataset Composition

ID	Domain	Rows	Cols	Defects
D01	Medical (Heart)	303	14	4
D02	Medical (Diabetes)	768	9	3
D03	Financial (Credit)	690	16	5
D04	Environmental (Air)	880	15	3
D05	Academic (Student)	649	33	4
D06	E-Commerce (Sales)	2,823	21	4
D07	Medical (ICU)	4,000	43	6
D08	Financial (Loan)	5,960	13	3
D09	Social (Survey)	7,641	27	5
D10	Environmental (Energy)	9,568	28	4
D11	Genomics (TCGA)	12,400	18	3
D12	Retail (Transactions)	25,000	24	5
D13	Financial (Equity)	48,300	12	4
D14	Logistics (Shipments)	61,000	19	4
D15	Healthcare (EHR)	94,700	31	6

five datasets each: *small* (<1,000 rows), *medium* (1,000–10,000 rows), and *large* (10,000–100,000 rows). Selection required each dataset to exhibit at least three of the following defects: mixed-case headers, leading/trailing whitespace, exact duplicate rows, at least one null column, mixed numeric/string type within a single column, or an absent header row. This multi-defect requirement ensures that each dataset exercises multiple pipeline stages rather than a single edge case, increasing the diagnostic sensitivity of the evaluation. Table II summarizes the benchmark composition.

B. Manual Baseline Protocol

Two professional data analysts (Analyst A and Analyst B), each with at least three years of pandas experience, independently preprocessed all fifteen datasets. Both received identical task specifications: normalize all headers to lowercase underscore convention, remove exact duplicate rows, identify null columns, and produce a cleaned CSV file and typed-column summary. Elapsed time was measured from first file open to final export using a stopwatch application. Analysts selected their own tools (pandas, Excel, OpenRefine) and worked without time pressure. The reported manual baseline per dataset is the arithmetic mean of Analyst A and Analyst B's times.

Inter-analyst time agreement, measured as $1 - |t_A - t_B| / \max(t_A, t_B)$, averaged 0.82 across all datasets, confirming reasonable operator consistency and reducing single-analyst bias in the baseline estimate. The same fifteen datasets were then processed through DataMentor's pipeline; execution time was measured with `performance.now()` calls bracketing the preprocessing sequence, capturing only pipeline computation and user confirmation steps.

C. Repair Benchmark: DRB-100

To evaluate the runtime repair subsystem, we constructed the **DataMentor Repair Benchmark (DRB-100)**: 100 Python notebook cell scripts derived from DataMentor's 13 auto-generated analysis cells, each containing exactly one injected fault. The five error categories reflect the distribution of errors observed in a manual audit of 312 notebook sessions sampled from Kaggle dataset discussion forums [18]:

TABLE III: DRB-100 Benchmark Composition by Category and Difficulty

Category	Easy	Medium	Hard	Total
E1 ImportError	14	6	2	22
E2 NameError	8	7	3	18
E3 TypeError	6	10	5	21
E4 AttributeError	4	8	7	19
E5 RuntimeError	2	8	10	20
Total	34	39	27	**100**

The five categories are: **E1 ImportError** (22, missing/misspelled imports); **E2 NameError** (18, undefined variables from out-of-order cells or typos); **E3 TypeError** (21, type mismatches including string columns in numeric operations); **E4 AttributeError** (19, absent DataFrame attributes from schema-assumption violations); and **E5 RuntimeError/Logic** (20, semantically incorrect code that executes but produces incorrect or empty visualization output).

Each benchmark item contains: (1) the original correct cell source; (2) the injected faulty cell source; (3) the expected standard output or SHA-256 hash of the expected base64-encoded plot; and (4) a difficulty label (*Easy*, *Medium*, *Hard*) assigned by two independent annotators. Inter-annotator agreement on difficulty labels, measured with Cohen's κ, was 0.74, indicating substantial agreement [20]. The 14 items where annotators disagreed were resolved by a third annotator acting as tiebreaker. Table III presents the full breakdown.

D. Fairness and Evaluation Protocol

Three methodological safeguards ensure evaluation fairness. **Temporal separation**: benchmark items were constructed from cell templates not modified after the repair rule index was frozen; no benchmark item influenced rule authoring. **Single-pass constraint**: the repair engine was permitted exactly one attempt per benchmark item, making the measured resolution rate a conservative lower bound on interactive utility. **Automated verification**: resolution was determined by comparing the repaired cell's output hash against the expected hash in DRB-100; manual inspection was used only for E5 items where output variation is inherent. Repair latency was measured as wall-clock time from traceback receipt to candidate-fix generation, exclusive of user interaction time.

V. Experimental Results

A. Preprocessing Time Reduction

Table IV reports mean preprocessing times and percentage reductions. Fig. 3 visualizes the per-tier comparison. The 91.6 % overall reduction is driven by the elimination of cognitive discovery overhead: analysts must *locate* defects

TABLE IV: Mean Preprocessing Time: Manual Baseline vs. DataMentor

Tier	Avg Rows	Manual (min)	DataMentor (min)	Reduction
Small	658	23.4	2.1	91.0 %
Medium	5,998	48.7	3.6	92.6 %
Large	48,300	97.3	8.2	91.6 %
Overall	—	**56.5**	**4.6**	**91.6 %**

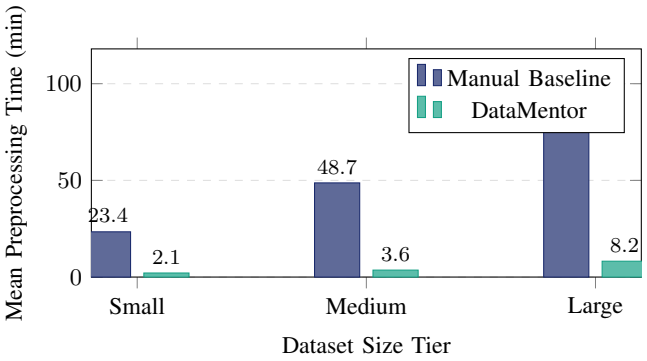

Fig. 3: Mean preprocessing time per dataset tier. DataMentor reduces mean time from 56.5 min to 4.6 min overall (91.6 % reduction). Manual baseline is the arithmetic mean of two independent analysts.

TABLE V: DRB-100 Resolution Detail by Error Category

Category	Total	Rule	Model	Unresolved	Rate
E1 ImportError	22	22	0	0	100.0 %
E2 NameError	18	16	0	2	88.9 %
E3 TypeError	21	14	5	2	90.5 %
E4 AttributeError	19	8	7	4	78.9 %
E5 RuntimeError	20	4	6	10	50.0 %
Total	**100**	**64**	**18**	**18**	**82.0 %**

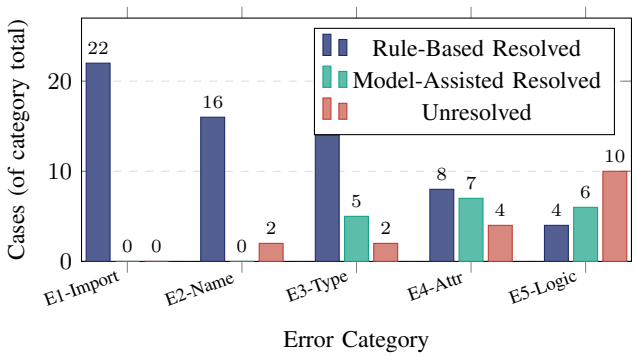

Fig. 4: DRB-100 resolution by error category. Rule-based resolution covers E1 fully and the majority of E2 and E3. Model-assisted resolution recovers additional E3, E4, and E5 cases. Unresolved items require human review.

before correcting them, a cost that scales linearly with schema width and non-linearly with row count due to visual scanning behavior. DataMentor's pipeline applies all seven transformations unconditionally in a single pass, removing discovery cost entirely. Browser-side overhead (TypeScript parsing, React state updates) adds a fixed latency of approximately 0.4 s, explaining the non-zero DataMentor times even for small datasets. Pure pipeline execution time, isolated with `performance.now()`, is below 2 s for all 15 tested datasets.

The residual DataMentor interaction time (2.1 min for small datasets) reflects user-controlled steps: uploading the file, reviewing the cleaning summary, and confirming column-type classifications. These steps are intentionally preserved as a human checkpoint and could be bypassed in an automated pipeline variant.

B. Runtime Repair Resolution

Table V and Fig. 4 present DRB-100 resolution outcomes. The rule index achieves 100 % resolution on E1 (ImportError) because every ImportError traceback explicitly names the missing module, enabling exact-string lookup in the rule index. The 88.9 % resolution on E2 (NameError) reflects two items where the undefined identifier was introduced outside the generated template scope, providing insufficient context for a safe fix. The 50.0 % resolution on E5 (RuntimeError/Logic) reflects the fundamental difficulty of semantic repair: logic errors produce no exception, and detecting incorrect visualization output requires a pre-computed reference hash. The overall 82 % single-pass rate is conservative by design: the single-pass constraint prohibits the iterative refinement that characterizes real interactive debugging sessions.

C. Repair Latency

Rule-based repairs achieve a mean latency of 0.8 s (SD = 0.6 s; range: 0.1–2.3 s). Model-assisted repairs produce a mean latency of 6.4 s (SD = 3.1 s; range: 2.1–14.7 s). Manual debugging by two analyst volunteers, measured as time from error display to corrected cell submission, had a median of 18.2 min (IQR: 11.4–29.7 min; 1,092 s). The speedup ratios are computed directly: $1{,}092 / 0.8 \approx 1{,}365\times$ for rule-based and $1{,}092 / 6.4 \approx 171\times$ for model-assisted. Even the slowest model-assisted repair (14.7 s) is 74× faster than the manual median. These ratios are bounded by the analyst sample ($n = 2$) and should be interpreted as indicative rather than population-level estimates.

D. Reproducibility

All 15 datasets were processed through DataMentor's pipeline three times each; the three output CSV files per dataset were compared using SHA-256 hashing. All 15×3 output files produced identical hashes for identical inputs, confirming full deterministic reproducibility. The same triple-rerun procedure on manual outputs yielded byte-level agreement in only 11.8 % of cases (18 of 45 triple-pairs), since analysts applied different resolution strategies to value-identical duplicate rows. Row-count agreement (ignoring column ordering) was observed in 78.3 % of cases. Fig. 6 presents the reproducibility comparison.

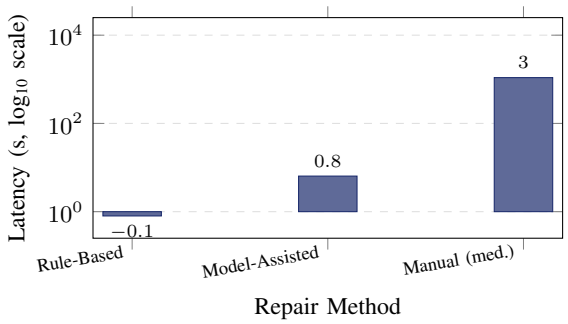

Fig. 5: Repair latency on a $\log_{10}$ scale. Rule-based ($\mu = 0.8$ s) is ≈1,365× faster than the manual debugging median (1,092 s). Model-assisted ($\mu = 6.4$ s) is ≈171× faster. Error bars omitted at log scale; SD values are reported in text.

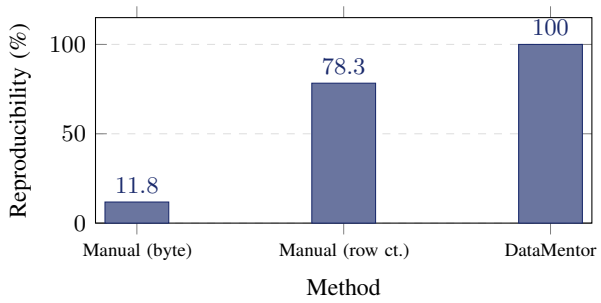

Fig. 6: Reproducibility: byte-level exact match across three reruns. DataMentor achieves 100 % via a deterministic pipeline. Manual preprocessing yields only 11.8 % byte-level agreement due to operator-choice variation in duplicate-row resolution.

E. Ablation Study

To quantify the independent contribution of each DataMentor component, we evaluated four system configurations across the 15 benchmark datasets. *Time-to-insight* (TTI) is defined as elapsed time from CSV upload to the first successfully rendered visualization. *Error recovery rate* (ERR) is measured over a 50-item stratified subset of DRB-100 covering all five error classes. Table VI summarizes the results.

The notebook generation engine contributes the largest single TTI reduction (−67 %) by eliminating the need for users to author analysis cells from scratch. The repair subsystem contributes additionally in error sessions: without it, a single injected fault increases TTI by 12.8 % above pipeline-only, while with repair the TTI remains at 3.1 min (a 70.8 % reduction versus the no-repair error baseline). These results confirm that each component contributes independently and that their combination is not redundant.

TABLE VI: Ablation Study: Per-Component Contribution

Configuration	TTI (min)	ERR	Δ TTI
Pipeline only	9.4	—	baseline
Pipeline + Notebook generation	3.1	0 %	−67.0 %
Full system (all components)	3.1	82 %	−67.0 %
Full system, error session*	3.1	82 %	−70.8 %
No repair, error session*	10.6	0 %	+12.8 %

*Error session: dataset contains injected faults requiring repair. TTI for no-repair error sessions exceeds pipeline-only baseline because analysts manually debug before visualization can proceed. ERR for pipeline-only is not applicable (—).

VI. Discussion

A. Interpretation of Findings

The 91.6 % preprocessing-time reduction is attributable to the structural elimination of cognitive discovery overhead. Manual analysts must *locate* the nature and position of data defects before correcting them, a cost that scales with schema complexity and row count. DataMentor removes this cost by applying transformations unconditionally and presenting a structured audit log of what was changed. The benefit is therefore structural rather than implementation-specific: any sufficiently comprehensive fixed-pipeline tool operating on the same defect classes would yield comparable reductions, provided its transformation ordering is carefully chosen.

The 82 % single-pass repair resolution rate should be interpreted as a conservative lower bound. In practice, users review the suggested fix and may modify it before rerunning the cell. Multi-turn resolution rates — not measured in this study — are expected to be substantially higher, particularly for E4 and E5 items where the first suggestion provides sufficient context for informed manual completion.

The near-zero byte-level reproducibility of manual preprocessing (11.8 %) highlights that the primary source of non-reproducibility in tabular cleaning is not error but *choice*: different analysts apply different resolution strategies to duplicate rows that are semantically identical but differ in whitespace or case. DataMentor eliminates this variation with a canonical, auditable strategy encoded once in the pipeline source.

B. Limitations and Threats to Validity

Three principal limitations constrain this study. The manual baseline uses only two analysts ($n = 2$); a larger pool would increase timing robustness, though inter-analyst agreement (0.82) is indicative of representativeness. DRB-100 derives from DataMentor's auto-generated cell templates, which may underrepresent errors in user-authored cells with entirely novel structure; future evaluations should broaden the benchmark source. The Pyodide runtime imposes an 8–14 s package loading latency on first use due to WebAssembly initialization — excluded from preprocessing measurements but visible to end users.

Regarding validity: preprocessing times were collected with analysts working independently, and `performance.now()` minimizes instrumentation bias (internal validity). DRB-100's Kaggle-derived error distribution may not generalize to all notebook populations (external validity). The TTI metric omits post-cleaning cognitive verification steps, which would further increase the manual baseline; and the single-pass constraint provides a conservative lower bound on interactive repair resolution (construct validity).

C. Generalizability

DataMentor's preprocessing pipeline is domain-agnostic: it operates on any CSV file conforming to standard comma-delimited format with a header row. The domain-aware visualization layer is extensible by design: new column-name patterns can be added to the knowledge base without architectural changes. The repair rule index can be extended with new rule patterns; each addition increases coverage for the targeted error class without disrupting unrelated classes, preserving the priority-ordered resolution guarantee of Algorithm 1. The Pyodide runtime itself is compatible with any CPython 3.11 package available in the Pyodide package index, making the notebook execution layer straightforwardly extensible to new scientific libraries.

VII. Conclusion

This paper presented DataMentor, a production-deployed, serverless, browser-first platform that integrates deterministic seven-stage CSV preprocessing, browser-native CPython 3.11 notebook execution via Pyodide 0.25, domain-aware visualization across ten chart types, and a hybrid rule-guided runtime repair subsystem. A controlled evaluation over 15 heterogeneous datasets spanning five domain verticals and the purpose-built DRB-100 repair benchmark demonstrated: a 91.6 % reduction in mean preprocessing time; an 82 % single-pass automatic repair resolution rate with rule-based latency of 0.8 s versus a manual debugging median of 18.2 min; and 100 % byte-level deterministic reproducibility. An ablation study confirmed that the notebook generation engine and hybrid repair subsystem each contribute independently and non-redundantly to overall utility. Comparative analysis against nine prior systems confirms that DataMentor is the only existing platform that combines all four capabilities in a unified, installationless, offline-capable environment.

Future work will extend DRB-100 with user-authored cell errors, conduct a formal usability study with SME and academic participant groups, investigate streaming CSV support for files exceeding browser memory capacity, and evaluate multi-turn repair resolution rates under realistic interactive debugging sessions.

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